

PROJECT DESIGN DOCUMENT

EMPLOYEE TRAINING EFFECTIVENESS DASHBOARD

Microsoft Power BI Business Intelligence Project

Project Type: Data Analytics / Business Intelligence

Platform: Microsoft Power BI

Dataset: 5,000 Employee Training Records

Professional Project Design Document

1. Document Control

Document Title	Employee Training Effectiveness Dashboard
Document Type	Project Design Document (PDD)
Project Domain	Employee Training / Learning & Development
Technology	Microsoft Power BI
Data Source Format	CSV / XLSX
Dataset Size	5,000 records, 10 columns
Primary Output	Interactive Power BI Dashboard

2. Executive Summary

The Employee Training Effectiveness Dashboard is a Power BI business intelligence solution developed to evaluate the effectiveness of employee training programs. The dashboard brings together training participation, completion status, pre- and post-training assessment scores, employee satisfaction, and training cost into an interactive reporting solution.

The project follows an end-to-end BI workflow: data preparation, modeling, calculated columns, DAX measures, visualization, interactive filtering, detailed analysis, and bookmark-based navigation.

3. Project Background

Organizations invest in employee training to improve skills and performance. However, training data can contain large numbers of records and multiple dimensions such as department, region, program, completion status, score, cost, and satisfaction. A dashboard can bring these indicators together so that training effectiveness can be analyzed in a structured way.

4. Problem Statement

The project addresses the need to understand whether employee training programs are producing measurable learning improvements while also monitoring completion, satisfaction, and expenditure. The dashboard is designed around the business question: “Are our training programs actually improving employee performance, and is the spend justified?”

5. Project Objectives

The main objectives are:

- Measure training ROI and learning improvement.
- Track training completion and drop-off.
- Identify completion gaps across departments, regions, and programs.
- Analyze employee satisfaction.
- Monitor training expenditure and cost per completion.
- Compare training programs using learning-gain measures.
- Provide an interactive dashboard for business analysis.

6. Business Questions

1. What percentage of enrolled employees completed their training?
2. How much do employee scores improve after training?
3. Which training programs show stronger average learning gains?
4. Which departments have higher or lower completion rates?
5. What is the overall employee satisfaction level?
6. How much has been spent on training?
7. What is the training cost per completed employee?
8. What proportion of employees dropped out?
9. How do training cost and score improvement relate?
10. How do results change when filtered by region, department, or enrollment period?

7. Project Scope

In Scope

- Training participation and completion analysis
- Score improvement analysis
- Satisfaction analysis
- Training cost analysis
- Department, region, and program comparisons
- Interactive dashboard filters
- Detail matrix and bookmark navigation

Out of Scope

The provided project material does not specify predictive modeling, automated forecasting, or external HR-system integration. These are therefore treated as future enhancements rather than current project deliverables.

8. Dataset Specification

The project dataset contains 5,000 employee training records and 10 columns. It includes 8 departments, 5 regions, 10 training programs, and approximately 24 months of enrollment data covering 2024–2025.

9. Data Dictionary

Field	Type	Description
EmployeeID	Text	Unique employee identifier
Department	Text	Sales, IT, HR, Finance, Operations, Marketing, Support, R&D;
Region	Text	North, South, East, West, Central
TrainingProgram	Text	One of 10 training courses
EnrollmentDate	Date	Employee enrollment date
CompletionStatus	Text	Completed / In Progress / Dropped
PreTrainingScore	Number	Assessment score before training, 0–100
PostTrainingScore	Number	Score after training
TrainingCost_INR	Number	Training cost per employee in INR
SatisfactionRating	Number	Post-training feedback score, 1–5

10. Data Preparation & Cleaning

The project guide specifies importing the CSV/XLSX through Power BI, checking data types, storing enrollment dates as Date values and scores/cost as numeric values, and handling blanks in PostTrainingScore during Power Query preparation.

11. Data Modeling

The project material supports a single flat Training table for the dashboard, with the option to split the model into Fact_Training and dimension tables. A Date table can also be created and marked as the official date table.

12. Calculated Columns

The calculated columns are row-level fields used for grouping, filtering, and analysis.

Score Improvement

```
Score Improvement =  
IF(  
    ISBLANK('Training'[PostTrainingScore]),  
    BLANK(),  
    'Training'[PostTrainingScore] - 'Training'[PreTrainingScore]  
)
```

Enrollment Month

```
Enrollment Month =  
FORMAT('Training'[EnrollmentDate], "MMM YYYY")
```

Outcome Flag

```
Outcome Flag =  
SWITCH(  
    'Training'[CompletionStatus],  
    "Completed", "On Track",  
    "Dropped", "At Risk",  
    "In Progress"  
)
```

Cost Bucket

```
Cost Bucket =  
IF('Training'[TrainingCost_INR] < 200, "Low",  
    IF('Training'[TrainingCost_INR] < 400, "Medium", "High"))
```

13. DAX Measures

The project uses reusable measures for dashboard KPIs and visual analysis.

Completion Rate %

```
Completion Rate % =  
DIVIDE(  
    CALCULATE(COUNTROWS('Training'), 'Training'[CompletionStatus]="Completed"),  
    COUNTROWS('Training')  
)
```

Avg Score Improvement

```
Avg Score Improvement = AVERAGE('Training'[Score Improvement])
```

Avg Satisfaction

```
Avg Satisfaction = AVERAGE('Training'[SatisfactionRating])
```

Total Training Cost

```
Total Training Cost = SUM('Training'[TrainingCost_INR])
```

Cost per Completion

```
Cost per Completion =  
DIVIDE([Total Training Cost],  
    CALCULATE(COUNTROWS('Training'), 'Training'[CompletionStatus]="Completed"))
```

Drop-off Rate %

```
Drop-off Rate % =  
DIVIDE(  
    CALCULATE(COUNTROWS('Training'), 'Training'[CompletionStatus]="Dropped"),  
    COUNTROWS('Training')  
)
```

14. KPI Definitions

- **Completion Rate %:** Completed training records divided by total training records.
- **Avg Score Improvement:** Average difference between post-training and pre-training scores for rows with a calculated improvement.
- **Avg Satisfaction:** Average satisfaction rating.
- **Total Training Cost:** Sum of training costs.
- **Cost per Completion:** Total training cost divided by completed training records.
- **Drop-off Rate %:** Dropped training records divided by total training records.

15. Dashboard Design

The dashboard is organized around headline KPIs, trend analysis, program comparison, status distribution, department comparison, cost-versus-improvement analysis, and detailed drill-down reporting.

16. Dashboard Components

Dashboard Component	Configuration	Purpose
KPI Cards	Completion Rate %, Avg Improvement, Avg Satisfaction, Total Cost	Headline performance indicators
Enrollment Trend	Enrollment Month + employee count	Track enrollment over time
Completion by Department	Department + Completion Rate %	Compare department completion
Improvement by Program	TrainingProgram + Avg Score Improvement	Compare learning gains
Status Breakdown	CompletionStatus + employee count	Show status mix
Cost vs Improvement	TrainingCost_INR + Score Improvement + Program	Compare cost and learning gain
Detail Matrix	Department → Region → TrainingProgram + key measures	Detailed analysis
Slicers	Region, Department, EnrollmentDate	Filter dashboard

17. Detail Matrix Design

The Detail Matrix is designed as a hierarchical analysis. Rows contain Department, Region, and TrainingProgram. Values contain Completion Rate %, Avg Score Improvement, Avg Satisfaction, and Total Training Cost. This provides a drill-down view from department to region and then to individual training programs.

18. Interactive Filters

Slicers are provided for Region, Department, and EnrollmentDate so users can focus the dashboard on a selected geography, department, or time period.

19. Bookmark Navigation

Bookmark navigation provides two dashboard states: **Home** for the overview dashboard and **Details** for the detailed matrix. Navigation buttons can be configured as VIEW DETAILS and BACK TO HOME. The project uses grouped and hidden visuals to control which dashboard elements are displayed in each bookmark state.

20. User Flow

User opens the Home dashboard → reviews KPI cards and charts → applies Region/Department/Date filters → selects VIEW DETAILS → reviews the Detail Matrix → selects BACK TO HOME to return to the overview.

21. Example Results & Analysis

The project guide provides example findings based on its generated dataset. These values should be replaced with the final values from the actual dashboard if the working dataset produces different results.

Overall completion rate: 77.1%.

Average score improvement: +21.7 points.

Average satisfaction: 4.07 / 5.

Total training spend: approximately ₹15.4 lakh.

The example analysis reports that Leadership Essentials and Cloud Computing Basics produced average gains of about 22 points, while Communication Skills was about 21.2 points. It also reports approximately 79% completion for R&D; and Finance and approximately 74% for HR. Dropouts had lower average satisfaction than completers in the example analysis.

22. Business Benefits

The dashboard provides a consolidated view of training effectiveness and enables users to monitor completion, learning improvement, satisfaction, cost, and program-level performance. It also supports detailed analysis by department and region.

23. Testing & Validation

The project should be validated by checking:

- Data types and missing values
- DAX calculation results
- Percentage and currency formatting
- Slicer interactions
- Chart-to-filter interactions
- Detail Matrix hierarchy
- Bookmark visibility states
- VIEW DETAILS and BACK TO HOME navigation
- Dashboard readability and layout

24. Project Limitations

The project material describes a dashboard based on the supplied training dataset. It does not specify external data integration, predictive analytics, or automated forecasting. Therefore, conclusions are limited to the available training records and the measures implemented in Power BI.

25. Future Enhancements

Potential future enhancements include:

- Add a dedicated Date dimension for richer time intelligence.
- Add additional ROI or benefit measures if business benefit data becomes available.
- Add automated refresh using an appropriate data source.
- Add role-based access if the dashboard is deployed to different user groups.
- Add additional drill-through pages for individual programs or departments.
- Add predictive analysis if sufficient historical data becomes available.

26. Technical Skills Demonstrated

- Microsoft Power BI
- Power Query
- DAX
- Calculated Columns
- Measures
- Data Modeling
- Data Cleaning
- KPI Development
- Interactive Visualizations
- Slicers
- Matrix / Drill-down Analysis
- Bookmark Navigation
- Business Storytelling

27. GitHub Repository Structure

```
Employee-Training-Effectiveness/  
  README.md  
  Dataset/  
    Employee_Training_Data.xlsx  
  PowerBI/  
    Employee_Training_Effectiveness.pbix  
  Screenshots/  
    Home_Dashboard.png  
    Detail_Matrix.png  
  Documentation/  
    Employee_Training_Effectiveness_PDD.pdf
```

28. Resume Project Description

Built an end-to-end Power BI dashboard analyzing 5,000 employee training records using Power Query, calculated columns, and DAX measures to track completion rate, score improvement, satisfaction, training cost, and training effectiveness.

29. Conclusion

The Employee Training Effectiveness Dashboard converts employee training data into an interactive business intelligence solution. The project demonstrates the complete workflow from data preparation and modeling to DAX calculations, visualization, detailed analysis, and dashboard navigation. It provides a structured way to monitor training completion, learning gains, satisfaction, and expenditure.

30. Reference Basis

This PDD is based on the supplied Employee Training Effectiveness Power BI project guide, including its dataset specification, calculated columns, DAX measures, dashboard design, and example analysis.